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Presentation:

Mr. Kiser's back hurts when he pushes his cart caused by the handle being too low. He bends down about a foot and a half, or in other words his body bends over 50 degrees. Bending over that far causes immense pain in his lower back, or sciatica. He states that it hurts the most when he pushes the cart far out from him. Pushing the cart far away causes his back to arch, which he says is what is hurtful. This pain is experienced during and after the action, he says "leaving me with bad tightness". One reason that he experiences pain is because of the previous injury of two slipped discs, but others at different heights and ages also feel pain when pushing the cart at an angle. He often pushes a heavy cart due to the ACT packets that he transports, so in doing that he experiences a more painful result.

Justification:

Mr. Kiser wants an extension to the cart so that he doesn't experience pain while or after using it. The efficiency and the success of this product puts the stakeholders at risk. Primary stakeholders include himself and the students he teaches. If the product malfunctions, Mr. Kiser might be late to class and also the students will be at risk of not getting the material they need. The design restricts the product to be over $30 to create and which causes the students to use mostly scraps from the robotics club. Restrictions are also that the product cannot be heavy or uncomfortable for the hands. The product also must be removable and aesthetically pleasing. The most important aspects are that the extension needs to be adjustable and it needs to be able to withstand all pressure applied to it.

Interview:

During the personal interview with Mr. Kiser, we obtained information about himself and the cart. While Mr. Kiser is 6 feet and 3 inches tall, he has a wingspan of about 75 inches. It is important to take into consideration both of these factors. The longer the wingspan is, the less he will have to bend over to push the cart. The cart that Mr. Kiser wants this product to work on has two shelves. Each shelf can hold about 250 pounds, holding around 500 pounds total. However, Mr. Kiser has a second cart that he sometimes uses which holds around 200 pounds each level, totaling 400 pounds. Kiser stated that the product can't be at a weight where the cart's ability to hold the same amount of weight as it does now is greatly changed. For the ACT test, he reaches the max weight that the cart can hold, so if the product is too heavy, then the efficiency of the cart will be hindered. Mr. Kiser specifically stated in the interview that he did not want the product to change the efficiency.